

significantly from the autonomous and GPS-independent nature of VECROS drones.

The startup designs and manufactures its own drones, including software and electronics development. Most component sourcing is localised within India, especially PCB boards and assembly processes. However, some raw materials and electronic components, such as ICs and resistors, are imported from countries like the US, Japan, Taiwan, and Australia. Its manufacturing and R&D facilities in Bengaluru are equipped with tools such as EMI/EMC testing systems, thrust stands, PCB soldering units, and oscilloscopes. VECROS currently produces approximately ten drones per month and maintains a lean team of 20 employees.

Athera is priced at approximately ₹2 million. Unlike conventional drones that require numerous skilled pilots, VECROS' AI-driven system reduces manpower needs, cutting both time and operational costs.

The startup claims to be strongly IP-driven, with approximately 15-20 patents filed and 10 granted, covering both design and method patents, as well as trademarks. Employees are encouraged to file patents, contributing to India's



Besta Prem Sai, Founder, VECROS Technologies

innovation ecosystem. The drones are intuitive to use and require minimal training, with user interfaces resembling gaming consoles. Simulators are provided to help users train before operating the actual systems.

The AI and GPU-based processing onboard reduces flight time from 40 to 30 minutes, but the added capabilities far outweigh this trade-off. The drones are resilient in extreme conditions, such as low light and fog, through the use of IR cameras and inertial measurement units (IMUs). Maintenance and calibration are rarely required, about once every 1000 flights, and the system automatically flags when recalibration is necessary.

Like most startups, VECROS has

faced significant challenges, particularly in securing funding. Indian VCs often evaluate startups based on revenue, making it difficult for deep-tech ventures with long-term R&D goals to attract capital.

For the future, VECROS plans to focus on developing drones for GPS- and radio-denied environments, which are becoming increasingly important in modern warfare. It aims to expand its team, raise more funding, and establish dedicated test facilities to refine and validate its systems.

The name 'VECROS' stands for 'Vector Robotic Operating System,' inspired by the physics concept of vectors—quantities that have both magnitude and direction. It reflects the ambition to build intelligent, directional, and purpose-driven robotic systems.

In terms of monetisation, the startup operates mostly in a service-based model for now, offering drone-based solutions such as 3D scanning and terrain modelling. Though it is currently pre-revenue in product sales, it generates about ₹400,000-500,000 per month through services. The startup is funded with ₹40 million in capital, which sustains its operations and R&D efforts.

—Nitisha Dubey

2 "WHY BUY A NEW KIT EVERY TIME WHEN ONE CAN DO IT ALL?"

From failed attempts to smart solutions, three friends turn hardware problems into a robotics kit. This could be the future path to learn AI and ML for school students as well as professionals.

In 2022, three college friends—Hitendra, Bhagyesh, and Guru—decided to start something new. Fascinated by robotics, especially autonomous robot navigation, they initially aimed to build indigenous industrial and customer-oriented robots to support industries in India. As they worked on their projects, they realised the real opportunity was in education, not just industrial

applications.

This year, in 2025, they registered their company as Ecruxbot Private Limited and shifted focus towards helping students learn robotics and advanced technologies. Their first product, Adhyay, is an educational kit designed for flexibility. Instead of buying a new kit for every project, students can swap development boards like

Arduino Nano, Raspberry Pi Pico, STM32, and ESP32 on the same custom motherboard. Adhyay allows students to begin with basic LED control and scale up to TinyML, IoT, and AI, running models directly on hardware with built-in sensors, Wi-Fi, and Bluetooth.

What makes it unique and how it helps its target users? Hitendra explains, "ESP32 supports both